Code No: 115EM **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2024
SOFTWARE ENGINEERING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What is CMMI?	[2]
b)	Describe the evolving role of software.	[3]
c)	Define feasibility study.	[2]
d)	Discuss about context models.	[3]
e)	What are component-level design elements?	[2]
f)	Write about architectural description languages.	[3]
g)	What is defect removal efficiency?	[2]
h)	Discuss about system testing.	[3]
i)	What is software quality?	[2]
j)	Describe about RMMM.	[3]

PART – B

(50 Marks)

2. Describe the importance of process model. Explain about specialized process models. [10]

OR

3.a) Explain about personal and team process models. [5+5]
b) Define software engineering. Explain about process patterns.

4. Draw state machine models of the control software for:
a) An automatic washing machine that has different programs for different types of clothes
b) The software for a DVD player
c) A telephone answering system that records incoming messages and displays the number of accepted messages on an LED. The system should allow the telephone customer to dial in from any location, type a sequence of numbers (identified as tones) and play the recorded messages. [10]

OR

5.a) Explain about the software requirements document.
b) Explain about behavioral system model. [5+5]

6.a) Describe user interface design patterns.
b) What are various types of coupling? Explain in detail. [5+5]

OR

7.a) Discuss about an architecture trade-off analysis method.
b) Explain about pattern-based software design. [5+5]

8. What is software testing? Why software construction is important? Explain about a strategic approach to software testing. [10]

OR

9.a) Explain about metrics for design model.
b) Discuss about metrics for software quality. [5+5]

10. Explain the following:
a) Risk strategies
b) Software reliability. [5+5]

OR

11. Explain the following:
a) SQA activities
b) Risk refinement. [5+5]

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